Intelligent Verification System for Concrete Compliance & QA Risk Management

BEAMP Cycle 6 Proposal — Ambient Systems

Ivan Damnjanovic, Founder & Lead Applicant

Section 1
About the Company

- **Company Overview**
- **Business Activities**
- **Products / Services**
- **Business Targets**
- **Key Team Members**

Company Overview

Global Presence

- AI-driven financial and sustainability platform with operations in Serbia, Canada, and Hong Kong
- Spin-off from Singapore (2021) as a spin-off from NUS.

patent (WO)
- Panasonic Deep Tech Challenge win
- Google runner-up





Our Proven Expertise

AI/NLP Pipeline

Advanced multi-source data ingestion and processing capabilities

Compliance Automation

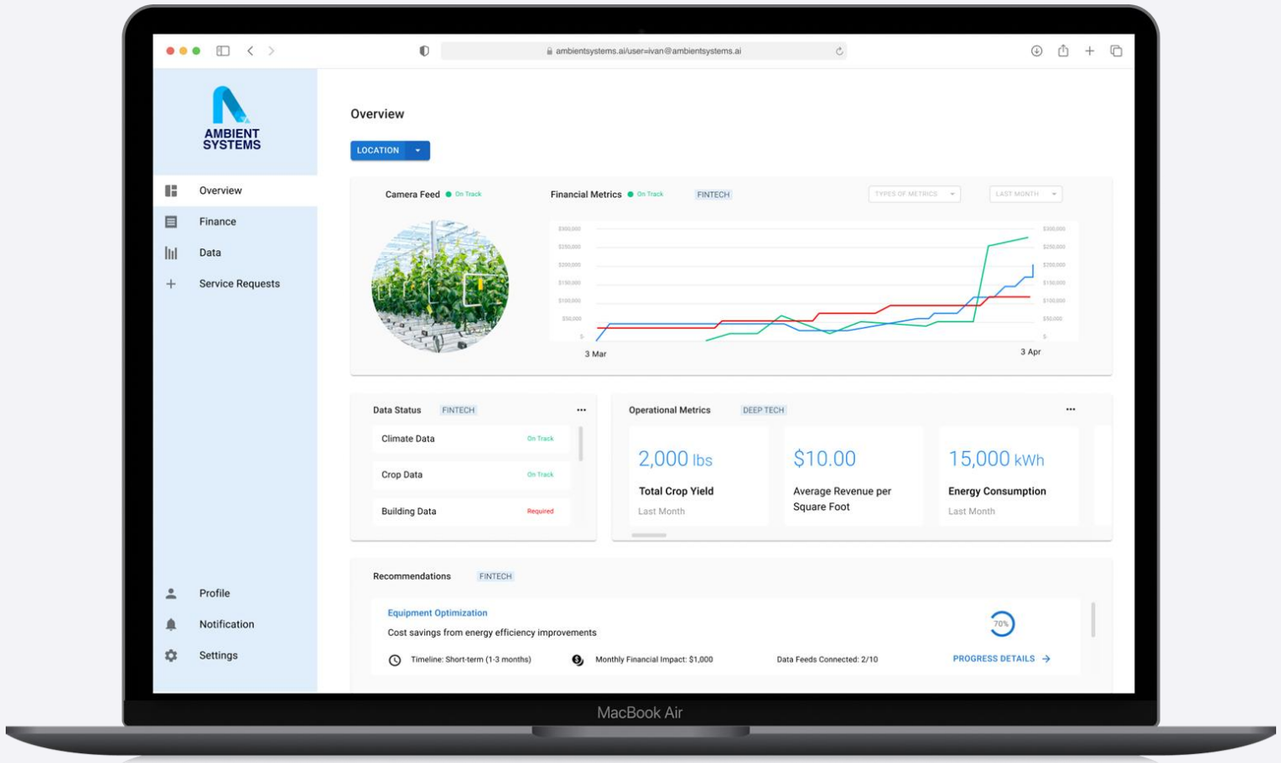
Specialized systems for building and equipment regulatory data

System Integrations

Seamless connectivity with PDFs, scans, WhatsApp, and supplier portals

Products/Services

Include your Ambient Twin and Ambient Verify modules (even conceptually) for real-time ESG and QA risk analytics.



POC/Pilot (BEAMP #6)

- Concrete quality
- Construction economics
- Sustainability metrics

Platform (ambientsystems.ai)

- Financial metrics
- Generative AI planning
- IoT connectivity

Consulting (ivandamnjanovic.com)

- Civil Engineering
- Energy systems
- Controlled Environment Agriculture (CEA)
- IP

Business Targets

Expansion into Singapore’s BEAMP ecosystem via FinTech/GreenTech/ConTech synergies;
aim to localize pilot for concrete QA and ESG integration.

- Ambient Systems is in the process of establishing a new Singapore business focused on applied AI for compliance, sustainability, and performance verification in the built environment.
- This new entity will leverage Ambient Systems’ prior R&D and IP portfolio (including WO) while operating as a dedicated Singapore-based venture to co-develop and pilot solutions with BEAMP ecosystem partners.
- Our prior operations across Canada, Hong Kong, and Serbia have validated the market need for integrated ESG-AI systems. The BEAMP program represents the ideal opportunity to anchor these capabilities in Singapore, supported by the country’s leadership in digital construction, standards adoption, and regulatory innovation.

Key Team Members

Myself (AI systems architect), and mention collaborators or advisors if available (can include academic or prior project ties).

Ivan Damnjanovic
Entrepreneur/Inventor
Serbia/HK

- *10 years in technical management, risk management, and AI*
- 5+ years in Asia (Singapore, Nepal, Hong Kong)
- Experience scaling **YC-backed startups**
- R&D Expert (NUS, NSERC, Centennial College)
- PhD-Level Expertise in Business Management and Economics
- Currently leading all development and strategic efforts
- Native & Fluent (Mother tongue) in Serbian and English
- Intermediate French (Reading and Writing)

Digital Market
Serbia/Norway

- *Advertisement*
- 5+ years in Asia (Singapore, Nepal, Hong Kong)
- Experience scaling **YC-backed startups**
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Planned Hires

- **Full-Stack Developer**
  - Responsible for platform architecture, AI feature builds, and product scaling

Section 2
Our Proposal

- **Summary**
- **Introduction**
- **Technology**
- **POC / Pilot**
- **Impact**
- **Scalability**



Summary

EXPAND
GROUP OF COMPANIES

Data Fragmentation

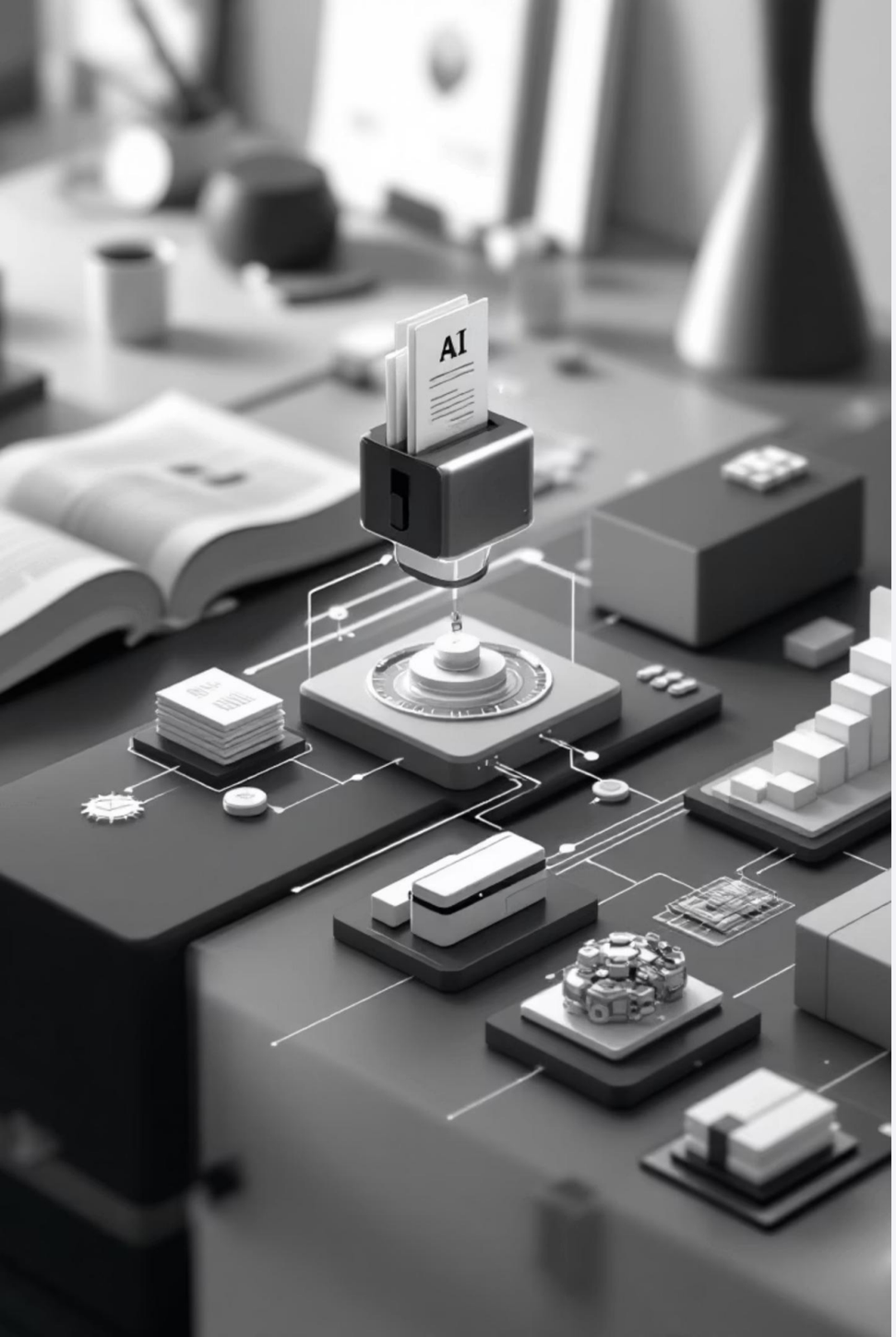
Concrete QA data scattered across paper delivery slips, PDFs, spreadsheets, and chat messages

Manual Processes

Time-intensive reconciliation leads to project delays, human errors, and compliance audit risks

Financial & Operational Risk

Finance teams approve milestone claims without comprehensive QA data verification



Introduction

Multi-Source Ingestion

Automated data extraction from diverse document formats

AI Processing Engine

Intelligent reconciliation and anomaly detection

Interactive Dashboards

Real-time insights for QA, finance, and audit teams

Technology



Universal Data Ingestion

Seamlessly processes PDFs, WhatsApp messages, electronic delivery orders, and spreadsheets from any source.



Intelligent Reconciliation

Automatically links delivery records, slump tests, and cube strength reports for complete traceability.



Smart Anomaly Detection

Configurable alerts flag missing data, non-conformance issues, and potential quality risks.



Role-Based Dashboards

Customized views for QA managers, finance teams, and audit compliance requirements.

POC/Pilot (BEAMP Opportunity)

Perfect Partnership

Expand Construction actively seeks automated concrete QA verification solutions for enhanced project oversight.

Strategic Benefits

BEAMP provides essential funding, real-world pilot environment, and ecosystem support for rapid development.

Cost Estimate

The approximate costs.

Deliverables & Milestones

Planned milestones.







Impact

70%

Reduction in Manual QA

Eliminate time-intensive data collation processes

100%

End-to-End Verification

Complete traceability from delivery through testing and approval

24/7

Audit Readiness

Continuous compliance monitoring and milestone claim validation

Transform fragmented QA data into structured datasets supporting ESG reporting and regulatory compliance.

Scalability

Technology Expansion

- Steel QA verification systems
- Prefabricated unit compliance
- BIM compliance automation

Business Model

SaaS platform with flexible pay-per-use API or enterprise deployment options.



Appendices

About the Company

- Item 1